Multimodal Cyberbullying Detection on Platform X Using Cross-Attention Fusion of IndoBERTweet and Vision Transformer

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***Abstract*—Cyberbullying on social media is a serious threat to victims' mental health, but most detection research for Indonesian content still uses text only. On platform X, the offensive meaning is often carried by the caption and the image together, so a post that looks harmless in text can become demeaning once the image is seen. This paper proposes a multimodal binary classifier that combines IndoBERTweet as a text encoder with a frozen Vision Transformer (ViT) as an image feature extractor, joined by a bidirectional cross-attention fusion in which text tokens act as queries and image patches as keys and values. The text branch is also enriched with a lexical TF-IDF feature that is added as one virtual token before fusion. We train and evaluate the model on 42,367 manually labeled Indonesian tweet-image pairs through five sequential experiments, from unimodal baselines to hyperparameter tuning and ensembling. The final ensemble of the three best cross-attention configurations reaches 86.44% accuracy, 87.61% precision, 81.97% recall, and 84.70% F1-score on the test set, higher than both unimodal baselines. Cross-attention performs as well as concatenation and element-wise multiplication, but unlike them it models the text-image interaction directly and keeps it easy to inspect.**

***Index Terms*—cyberbullying detection, multimodal fusion, cross-attention, IndoBERTweet, vision transformer**

I. Introduction

Cyberbullying is a recurring problem on Indonesian social media. Platform X (formerly Twitter) carries thousands of daily posts with insults, threats, or ridicule aimed at specific people [1]. Surveys report that hate speech and cyberbullying are a growing concern on Indonesian platforms, while detection tools for the language remain scarce [1], [15], [17]. The problem is harder now because abuse is not only text: users often attach images, memes, or screenshots that carry the insult, so a post with an ordinary caption can still be bullying when its image is offensive. Because the number of posts is very large, manual moderation does not scale, and a fast and measurable automatic classifier is needed.

Work on this problem for Indonesian is still limited. Surveys note that most local hate-speech studies use classical machine learning on text only [1]. For example, a hybrid CNN-BiLSTM with FastText expansion reaches 80.55% accuracy on an Indonesian cyberbullying dataset but uses no visual information [2]. Abroad, multimodal models are more advanced: Hate-CLIPper models text-image interaction through a feature interaction matrix and beats human performance on the Hateful Memes Challenge [3], and other studies build multimodal networks for cyberbullying [4]. These systems, however, are tested on English data, and none targets Indonesian.

We address this gap. Our model combines IndoBERTweet [5] for Indonesian tweet text with a Vision Transformer [6] that reads the image as patches, and links them through cross-attention where text tokens are queries and image patches are keys and values. To strengthen the text side, a TF-IDF feature selected by chi-square [7] is added to the token sequence as one virtual token before fusion, so lexical cues that are common in Indonesian cyberbullying are used together with the contextual representation. Unlike Hate-CLIPper [3], which builds on CLIP features for English memes, our approach targets Indonesian tweets, keeps the ViT frozen for efficiency under limited GPU memory, and injects an explicit lexical signal; to our knowledge this combination has not been applied to Indonesian multimodal cyberbullying.

We study three questions: (RQ1) which configuration of the IndoBERTweet-ViT cross-attention model works best; (RQ2) how cross-attention compares with concatenation and element-wise multiplication; and (RQ3) why the best model still makes mistakes. Table I links each experiment (labeled SK1–SK5, where SK means *scenario*, plus a final error-analysis stage) to these questions. Our contributions are: (i) a cross-attention multimodal cyberbullying detector for Indonesian platform X, a setting not covered by earlier work; (ii) a TF-IDF virtual-token injection that adds a lexical signal to the contextual encoder; and (iii) a transparent, scenario-based evaluation with an error analysis of the remaining mistakes. Table II states the gap in measurable terms.

II. Related Work

Kusuma and Chowanda [8] combined IndoBERTweet with BiLSTM for Indonesian hate-speech detection and found that IndoBERTweet beats generic BERT on informal text, but they used text only. Chhabra and Vishwakarma [9] used multi-scale

TABLE I
MAPPING OF EXPERIMENTS TO RESEARCH OBJECTIVES

Experiment	Objective	Answers
SK1	Data split and unimodal baselines	RQ1
SK2	Best text extraction configuration	RQ1
SK3	Contribution of TF-IDF injection	RQ1
SK4	Multimodal fusion method	RQ2
SK5	Final model (tuning and ensemble)	RQ1
Error analysis	False positive/negative analysis	RQ3

visual kernels with knowledge distillation for multimodal hate speech and improved macro-F1 on MAMI and MMHS150K, but on English data. Yang et al. [10] transferred knowledge between hate-speech and sarcasm detection, and Wu et al. [11] built the Fuser framework that weighs each modality adaptively; neither was tested on Indonesian. Recent work follows the same trend: Hafiza and Setiawan [12] combined IndoBERT with TF-IDF and FastText expansion in a hybrid CNN-LSTM, Aditya and Syarif [13] paired IndoBERTweet with BiLSTM, and Zhang et al. [14] showed a cross-platform multimodal method; the first two use text only, and the last is again non-Indonesian.

Table II compares these studies on five features. The Indonesian studies [2], [8], [12], [13] use text only, and each one is missing at least one feature: [2] adds FastText expansion but has no contextual encoder; [8] and [13] use a contextual encoder but no lexical expansion; and [12] has both but stays unimodal. The multimodal studies [3], [4], [9] work on English and rarely combine lexical expansion with explicit cross-modal attention. Our method is the only one with all five features, which supports our claim of a more complete approach for Indonesian content on platform X.

For background, TF-IDF [7] scores a term by how well it separates documents; we fit the vectorizer on the training data only to avoid leakage and keep 512 terms using a chi-square test. FastText expansion [2] fills in the sparse TF-IDF vector using sub-word (character n-gram) embeddings, which handle out-of-vocabulary words and spelling variants that are common in informal Indonesian. Cross-attention [18] lets one modality read the useful parts of another by taking queries from one side and keys and values from the other, which works well for detailed vision-language interaction.

III. PROPOSED METHOD

A. Dataset and Preprocessing

The dataset contains 42,367 Indonesian tweet-image pairs crawled from platform X between February and April 2026, covering posts published from 2022 to 2026. Each record has the tweet text, its accompanying image, and a binary label. Because no labeled Indonesian multimodal cyberbullying dataset was available, three annotators (the first author and two peers) labeled every pair manually as cyberbullying (1) or safe (0). The label is assigned at the post level, judging the tweet text *together with* its image, so that a caption that becomes offensive only in the presence of its picture is labeled correctly; this matches the multimodal nature of the task. Pairs without a usable image or with non-Indonesian text are excluded. The resulting set is roughly balanced (about 55% safe and 45% cyberbullying), and we split it with stratified sampling at 80:10:10 to preserve the label ratio in every partition.

Text is cleaned by replacing URLs with `HTTPURL` and mentions with `@USER`, removing the hash symbol while keeping the hashtag word, and stripping non-ASCII characters such as emojis. Non-standard words are normalized with a manually built 108-entry slang dictionary, and the text is tokenized with the IndoBERTweet tokenizer at a maximum length of 64. Each image is converted to RGB, resized to 224×224, pixel-normalized, and passed through the frozen ViT to obtain 196 patch embeddings.

B. Model Architecture

In the text branch, IndoBERTweet is the only source of word embeddings and outputs a 768-dimensional vector per token. In parallel, a TF-IDF vector is computed over a vocabulary fit only on the training data; the 512 most discriminative terms are kept with a chi-square test, and the sparse vector is enriched by FastText expansion, where empty entries are filled with the similarity scores of nearby words. This 512-dimensional lexical vector is projected by a linear layer into a single 768-dimensional virtual token and prepended to the BERT token sequence, giving a 65-token text sequence in which the lexical cue joins fusion without replacing the semantic representation. In the image branch, the frozen ViT-Base/16 produces 196 patch embeddings of size 768, used as keys and values.

Fusion uses bidirectional multi-head cross-attention with eight heads. In one direction the text tokens are the queries and the image patches are the keys and values; in the other direction the roles are swapped. Each block adds a residual connection and layer normalization. The two attended sequences are mean-pooled and concatenated into a 1536-dimensional vector, which a fully connected layer maps to the two classes. Fig. 1 shows the architecture. We emphasize that model selection at every stage uses validation F1 only; the test set is evaluated a single time for each winning configuration and is reported in Tables III and IV purely for transparency, never for tuning or model development.

We run five linked experiments, SK1–SK5, and end with an error-analysis stage. SK1 sets the split ratio and the unimodal baselines; SK2 finds the best pooling and maximum token length and the best TF-IDF setting; SK3 adds the FastText-enriched TF-IDF feature; SK4 fuses the two modalities with cross-attention; and SK5 tunes the cross-attention hyperparameters with Optuna (Tree-structured Parzen Estimator) and grid search, then ensembles the best runs. In every experiment we pick the winner by validation F1.

IV. RESULTS AND DISCUSSION

In SK1, the 80/10/10 split gave the best validation F1 (85.33%). The two modalities were very different: text reached about 85% validation F1 while image reached only about 65%, which shows that the cyberbullying signal here is mostly lexical and semantic. SK2 chose CLS pooling with a maximum

TABLE II
CAPABILITY COMPARISON OF REPRESENTATIVE STUDIES (✓ INDICATES THE CAPABILITY IS PROVIDED)

Study	Indonesian	Multimodal	Lexical expansion	Contextual encoder	Cross-modal attention
Nasution & Setiawan [2]	✓	–	✓	–	–
Kusuma & Chowanda [8]	✓	–	–	✓	–
Hafiza & Setiawan [12]	✓	–	✓	✓	–
Aditya & Syarif [13]	✓	–	–	✓	–
Kumar & Nandakumar [3]	–	✓	–	✓	✓
Chhabra & Vishwakarma [9]	–	✓	–	✓	–
Tabassum & Nunavath [4]	–	✓	–	✓	–
Proposed method	✓	✓	✓	✓	✓

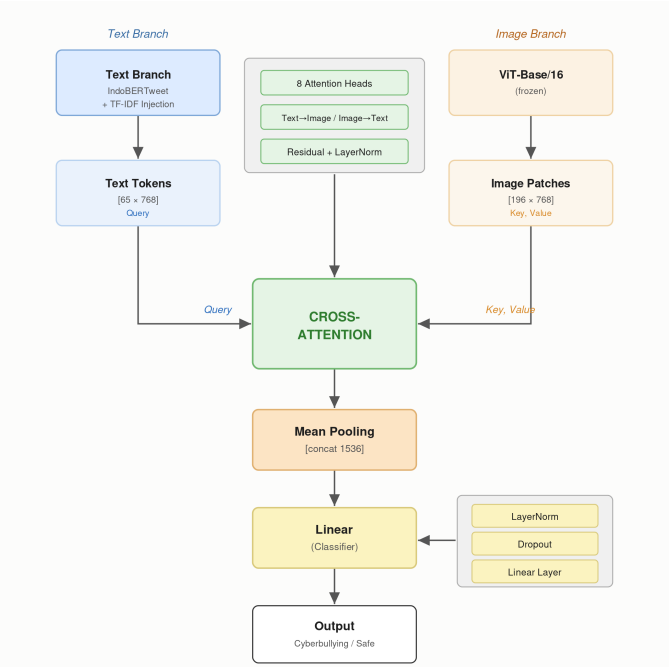


Fig. 1. Proposed multimodal architecture: cross-attention fusion of IndoBERTweet (text, query) and a frozen ViT (image, key/value), with a TF-IDF virtual token added to the text branch.

length of 64 (85.47%) and a TF-IDF setting with 10,000 features and 1–3 grams (85.71%). SK3 gave a best injection variant at 85.80% validation F1.

For RQ2, the three fusion methods in SK4 were close, as Table III shows: element-wise multiplication, cross-attention, and concatenation are within 0.6 point of validation F1. Although multiplication is marginally ahead (0.13 point), the three methods are effectively tied, so we select cross-attention for its ability to model and expose the text-image interaction through interpretable attention weights, which the passive fusions lack. A separate ablation confirmed the TF-IDF gain: virtual-token injection 85.82%, pooled injection 85.38%, and no injection 85.27% validation F1. SK5 gave the final model, an ensemble of the three best cross-attention runs, with 86.14% validation F1 and 84.70% test F1. Table IV lists the full chain, where validation F1 rises step by step from SK1 to SK5.

Each component adds a small but consistent gain along the chain (Table IV): the TF-IDF virtual token adds up to 0.55 point over the semantic baseline, fusion brings in the

TABLE III
FUSION METHOD COMPARISON (SK4)

Fusion method	Val F1	Test F1
Element-wise multiplication	85.95%	84.34%
Cross-attention (selected)	85.82%	84.12%
Concatenation	85.38%	84.66%

TABLE IV
SCENARIO CHAIN SK1–SK5 (WINNING MODEL)

Exp.	Winner	Val F1	Test F1
SK1	Text 80/10/10	85.33%	84.63%
SK2	CLS/64	85.47%	84.02%
SK2B	mf10k_unibitri	85.71%	84.62%
SK3	news_top3	85.80%	83.53%
SK4	CA-1-virtual	85.82%	84.12%
SK5	Ensemble (3 models)	86.14%	84.70%

image evidence, and ensembling stabilizes the result. The multimodal model beats both unimodal baselines: clearly over image and slightly over the best text-only model (Table V). The small gain over text is expected because the signal here is mostly lexical, so the text encoder is already strong. The image modality, although weak on its own (about 65% F1, still well above the 50% chance level, so ViT does capture some signal), helps mainly on borderline posts where the text is ambiguous but the image is clearly offensive.

On the test set, the final model correctly labels 2,074 safe and 1,591 cyberbullying posts, with 225 false positives and 350 false negatives out of 4,240 samples (Table VI). The higher number of false negatives shows up as a recall lower than precision. The error analysis shows the model leans on clear lexical cues: false positives happen when strong words are used in friendly jokes, and false negatives happen in indirect bullying that uses hints without any harsh word. We also see a complementary effect: the news-corpus expansion is the weakest on its own but the best once fused with IndoBERTweet, which shows that the value of an extra feature does not come from its strength alone.

A. *Limitations*

Three limitations temper these results. First, the multimodal gain over the strong text-only model is small, and our evaluation uses a single stratified split and a single training run; repeated runs and significance testing are needed to confirm that the gain is robust. Second, the image modality is comparatively

TABLE V
UNIMODAL VERSUS MULTIMODAL

Approach	Val F1	Test F1
Unimodal Image (frozen ViT)	65.65%	65.06%
Unimodal Text (IndoBERTweet)	85.80%	83.53%
Multimodal (Cross-Attention)	86.14%	84.70%

TABLE VI
CONFUSION MATRIX OF THE FINAL MODEL ON THE TEST SET

Actual \ Predicted	Cyberbullying	Safe
Cyberbullying	1,591 (TP)	350 (FN)
Safe	225 (FP)	2,074 (TN)

weak, partly because the frozen ViT is not adapted to social-media imagery and cannot read text embedded in memes. Third, the dataset comes from a single platform and time window and is labeled by three annotators, so generalization to other platforms, periods, and annotation schemes remains to be tested.

V. CONCLUSION

We presented a multimodal cyberbullying detector for Indonesian platform X that fuses IndoBERTweet and a frozen Vision Transformer with cross-attention and adds a TF-IDF virtual token to the text branch. Over five linked experiments, validation F1 grew from 85.33% to 86.14%, and the final ensemble reached 86.44% accuracy and 84.70% F1 on the test set, better than both unimodal baselines. Cross-attention matched concatenation and multiplication while modeling the cross-modal interaction directly, and the capability comparison shows that our approach is more complete than earlier Indonesian and multimodal work. Future work includes strengthening the weak image branch through limited ViT fine-tuning or OCR for text inside memes, using expansion corpora closer to the tweet style, adding statistical significance testing over repeated runs, and testing the model on other platforms and time periods.

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